

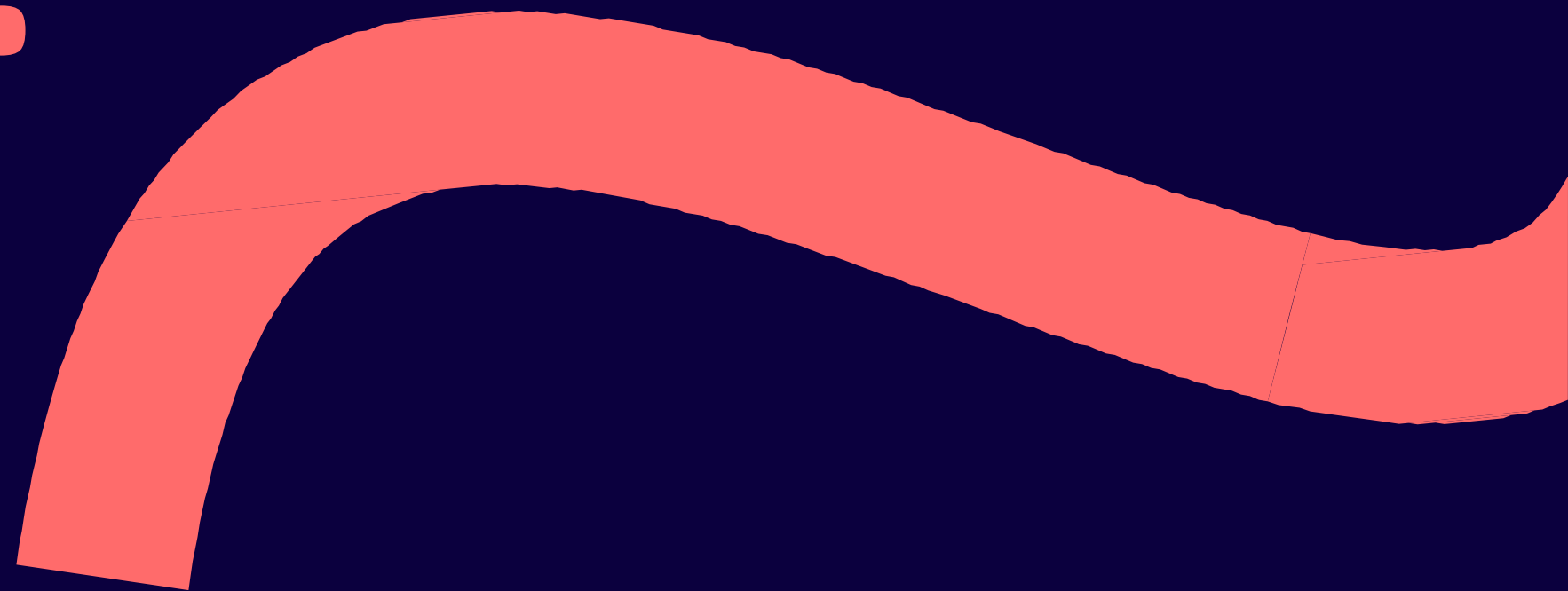


Infobip Shift Conference 2026

13-15 September 2026; Zadar, Croatia

PUBLIC

REST Delivers. MCP Operates.



Edvin
Teskeredžić

Senior AI Software Engineer
Infobip

ACT I:

INTRO

What I do

- AI and ML (obviously)
- Internal Tooling
- Developer Experience
- API Design

What I do

- AI and ML (obviously)
- Internal Tooling
- Developer Experience

- API Design

I <3 FastAPI



**FastAPI**

101,761 followers

4mo • 🌐

Here's FastAPI 0.135.3, adding support for fully vibe-coded apps ✨

Use the new `app.vibe()` decorator, embrace the AI future 🤖

Read the new docs: <https://lnkd.in/daXuutVx>

```
1 @app.vibe(  
2     "/vibe/",  
3     prompt="pls return json of users from database. make no mistakes",  
4 )  
5 async def ai_vibes(body: Any): ...  
6
```



[Redacted name] 
@ [Redacted handle]

Follow



RESTful APIs may be dead soon. Instead, web services may expose a single POST entry point for a prompt. Internally, an AI agent may decide how to interpret it and what to do with the data and the database.

6:31 PM · May 15, 2026 · **1M** Views



815



391



531



253



REST for the delivery path.
MCP for the operations
around it!

Why killing REST is seductive...

- One endpoint
- No SDKs
- No versioned docs
- The agent decides what to do



... and why we can't do it

- Unclear contracts
- Impossible to test
- Auth & observability nightmare
- Latency
- Cost

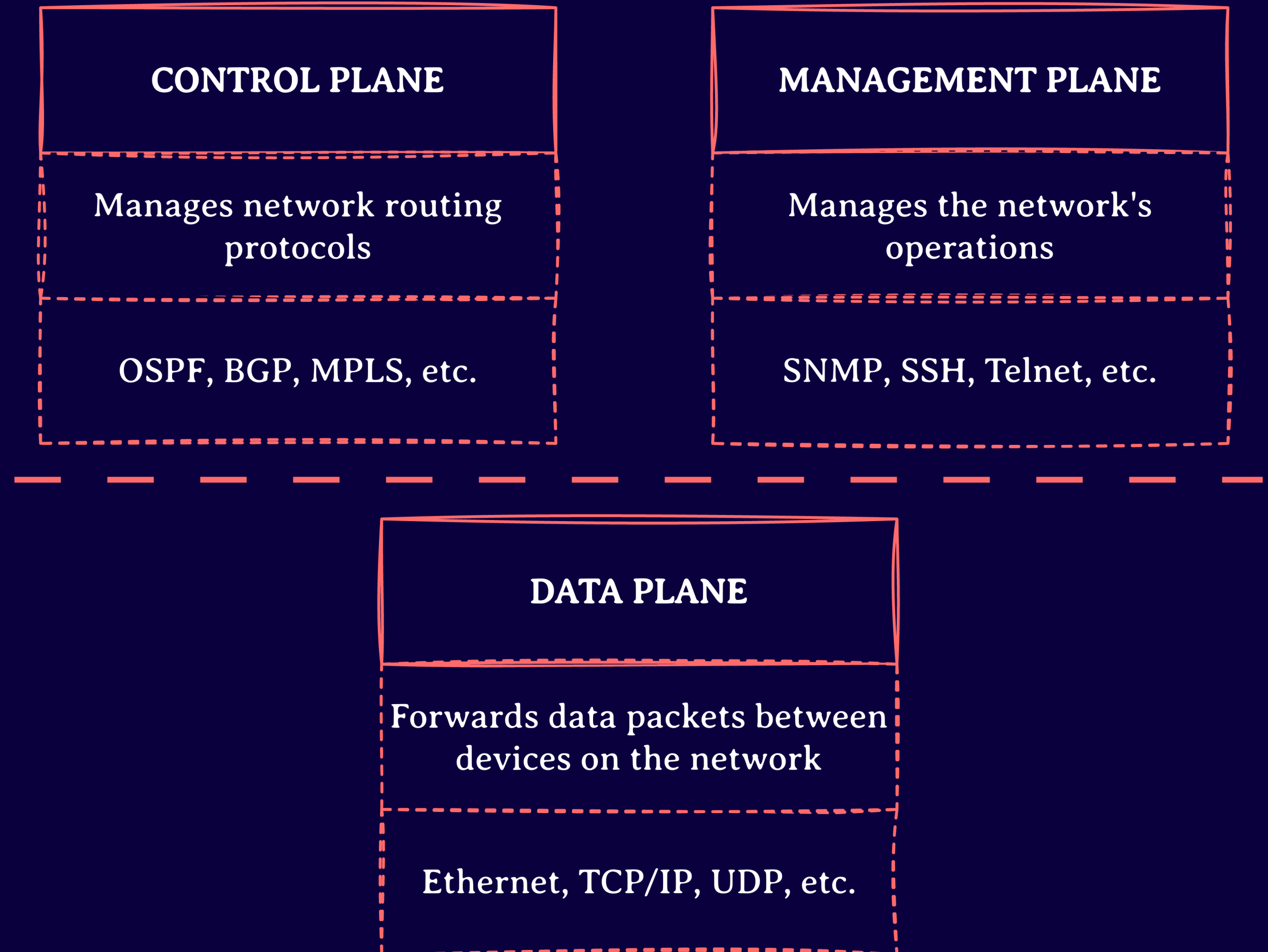


*Would you let an LLM
handle deterministic, high-
volume operations?*

ACT II:

THE MENTAL MODEL

**Borrowed from
people who
move packets**



Telco points the way

- They solved this problem 20 years ago
- Router forwards packets; a separate brain handles operations/routing
- We do not run BGP for every packet

Telco points the way

When the higher-level
planes fail, the data plane
must keep going!

How this translates

High-level planes

→ Agent/MCP

- Low-volume
- Latency-tolerant
- Decision-heavy
- High blast radius

Data Plane

→ REST delivery API

- Per-message
- Hot path
- Deterministic
- High volume

An easy way to remember

**MCP is not the thing.
MCP is the thing that gets
us to the thing!**

The contract is the product

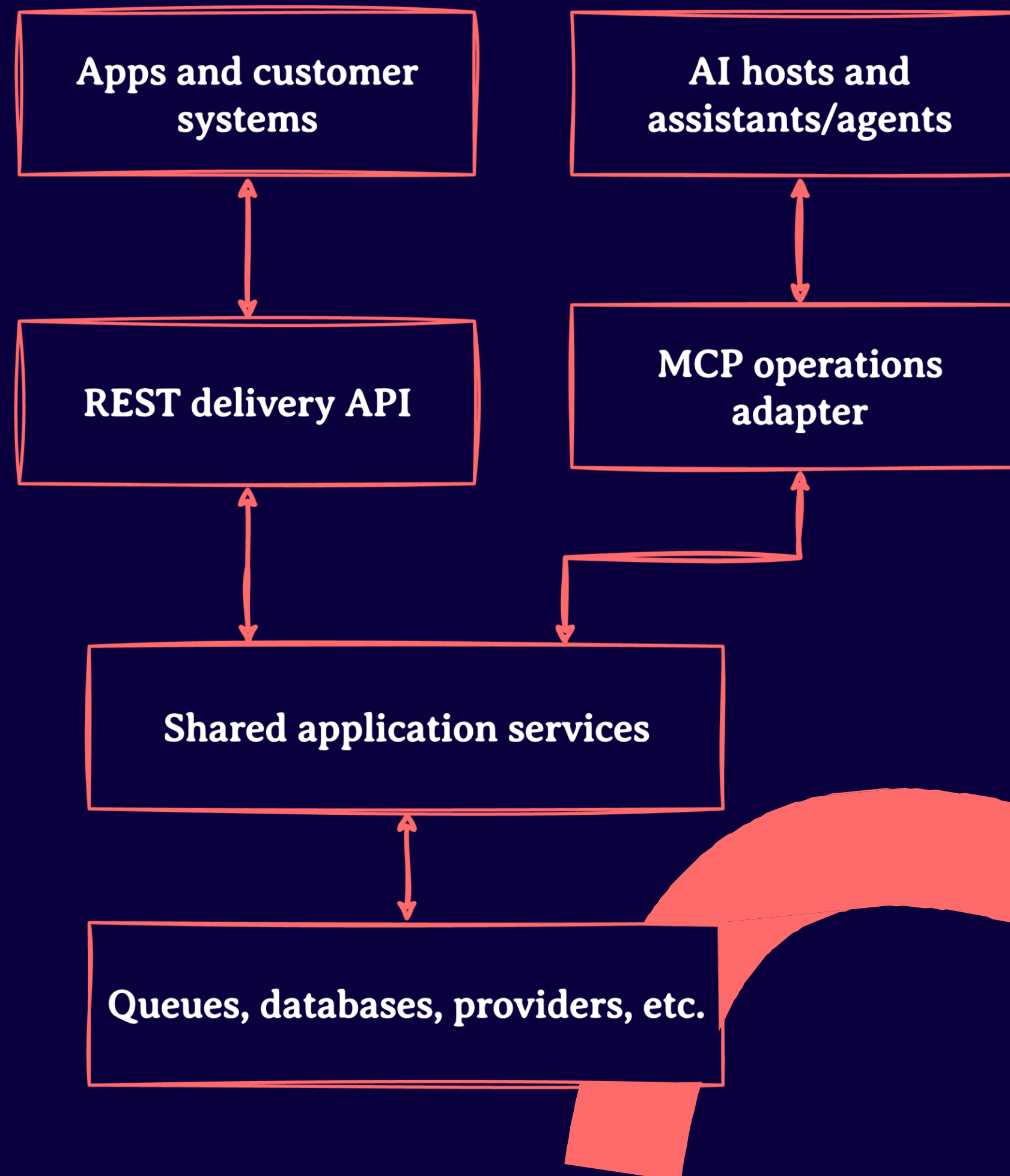
Boring + Deterministic + Testable + Repeatable



```
1  
2 POST /messages  
3   Idempotency-Key: 5f3a...-once  
4   { "to": "+385...", "channel": "sms", "text": "Your code is 123456" }  
5  
6   200 OK { "messageId": "abc", "status": "PENDING" }
```



MCP helps operate the product



ACT III:

MCP 101

MCP 101

- AI: The Mastermind – confident but also useless without a crew!
- The crew knows different domains
- **MCP gives them a shared briefing format**



MCP 101

MCP does not make the Mastermind smarter. It gives it access to a crew!





```
1 // context-gathering (read)
2 {
3     "name": "explain_failed_delivery",
4     "description": "Aggregate logs for one message and explain why it failed.",
5     "inputSchema": {
6         "messageId": "string"
7     }
8 }
9 // provisioning (write) – governed, not forbidden
10 {
11     "name": "provision_10dlc_number",
12     "description": "Register a US 10DLC number...",
13     "inputSchema": {
14         "brandId": "string",
15         "campaignUseCase": "string",
16         "areaCode": "string"
17     },
18     "annotations": {
19         "readOnlyHint": false,
20         "destructiveHint": false,
21         "idempotentHint": true
22     }
23 }
```



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```

These hints are
only advice!



The six good jobs for MCP

- Discovery
- Setup
- Operations/State
- Troubleshooting
- Guidance
- Safe Automation

These are ops, not delivery

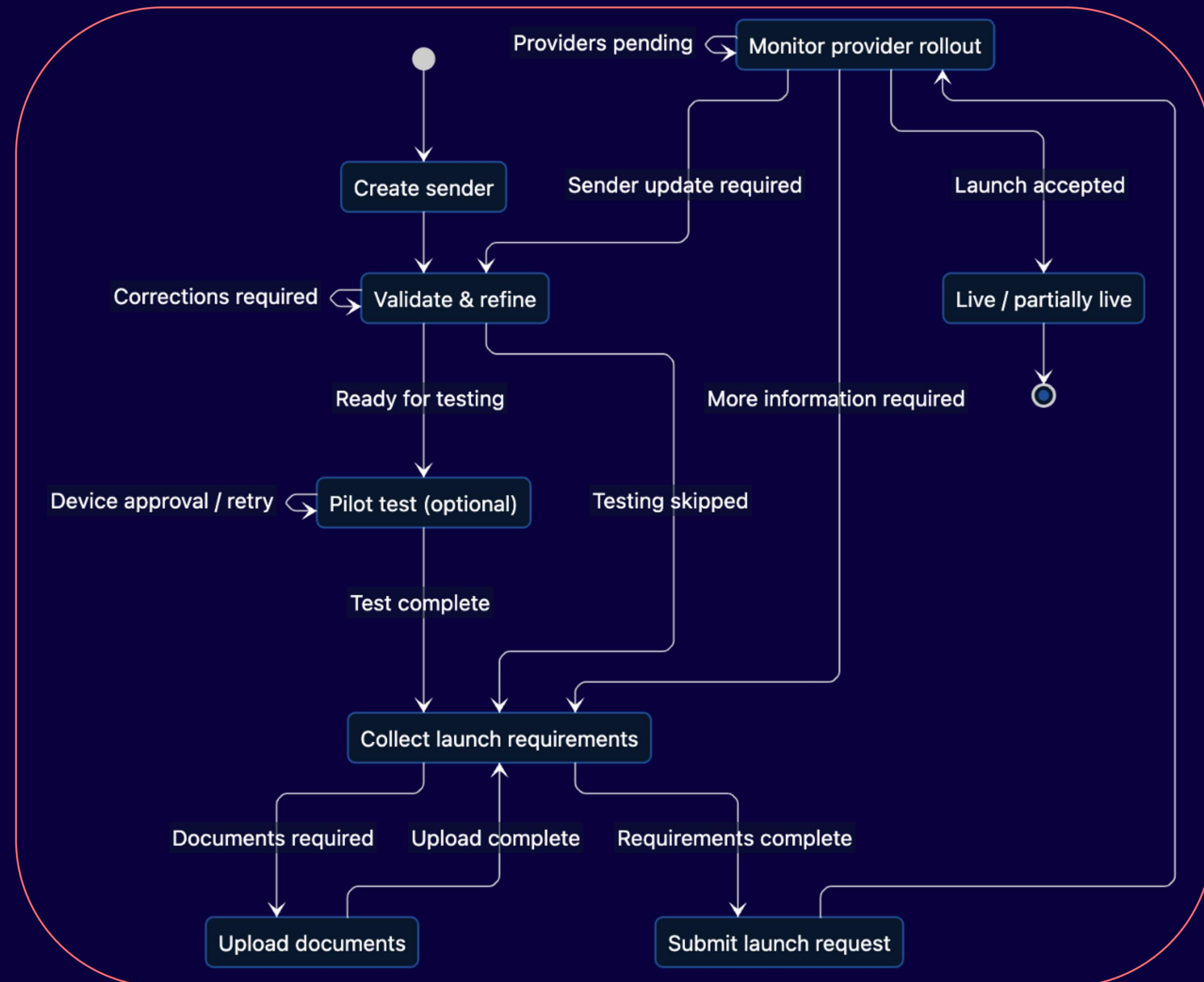
MCP or REST

TASK	REST	MCP	WHY
Send a message	✓	✗	Stable contract + latency
Receive delivery feedback	✓	✗	Event-driven, deterministic
Configure webhook	maybe	✓	Operational workflow
Explain a failed delivery	maybe	✓	Context aggregation
Register a sender	maybe	✓	Complex, multi-step flow
Generate an integration	✗	✓	Developer assistance
Bulk prod migration	✓	💀	Needs guardrails + audit

A perfect job for MCP

Something that is
genuinely miserable to
do by hand:

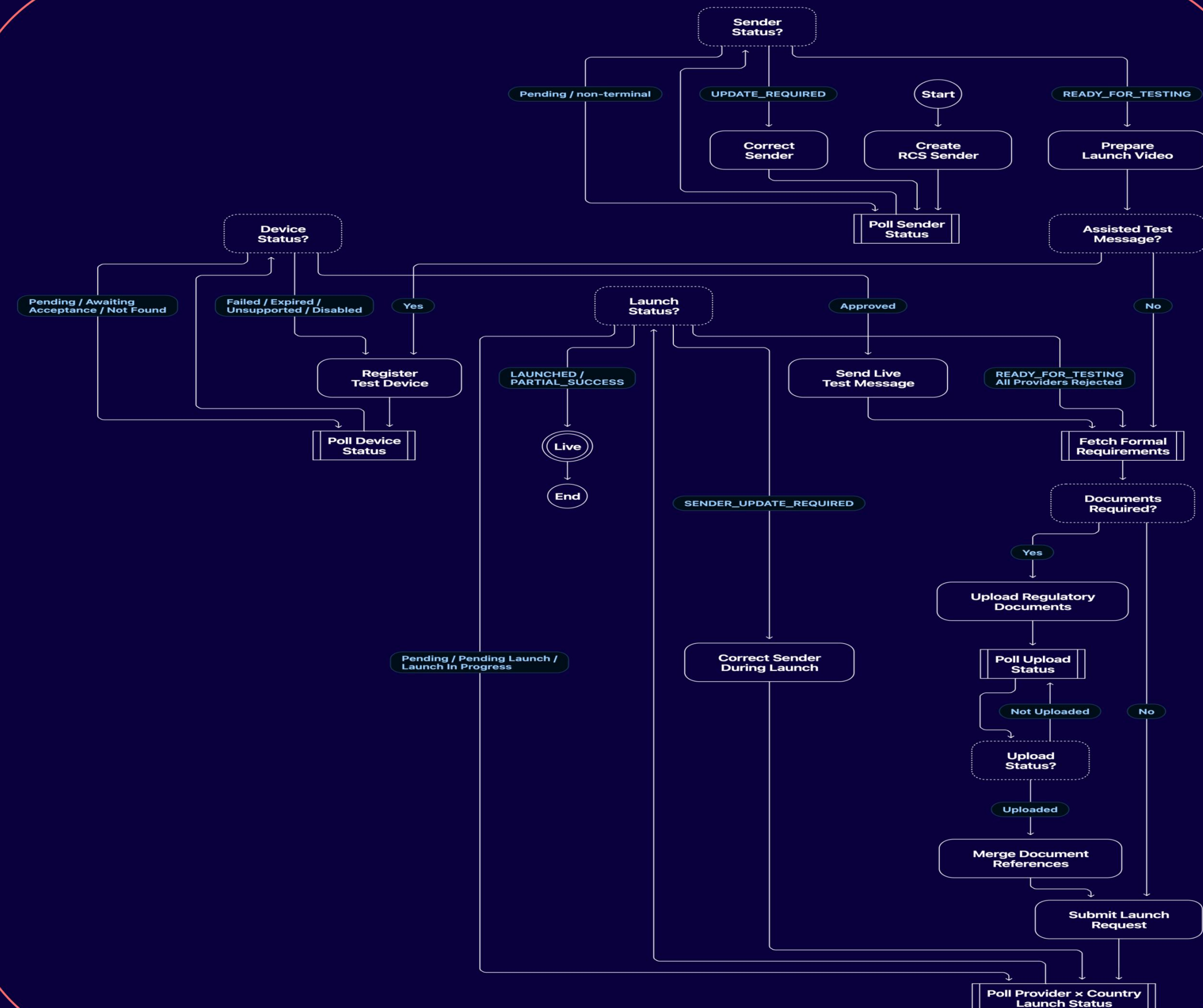
Sender Provisioning



A perfect job for MCP

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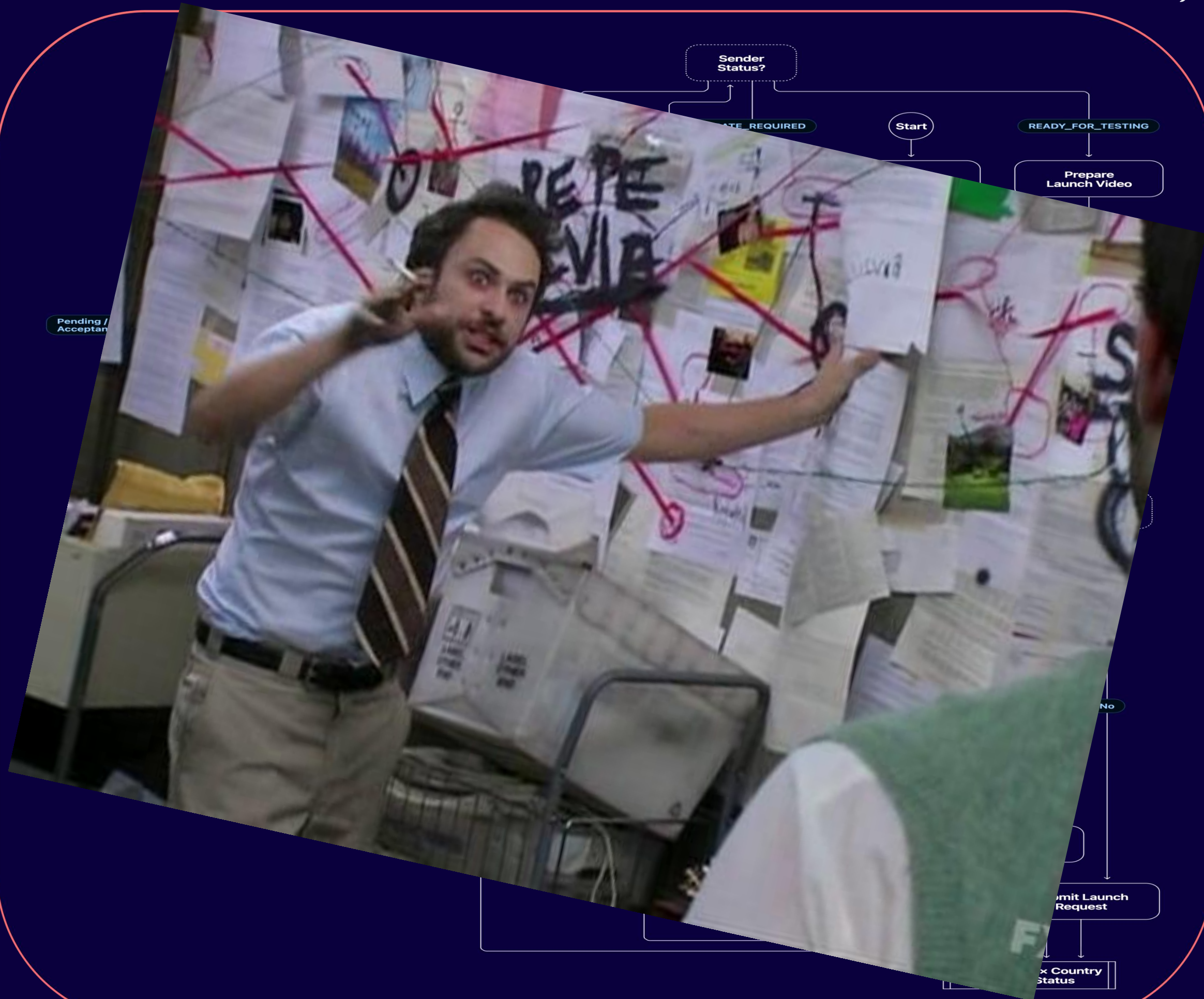
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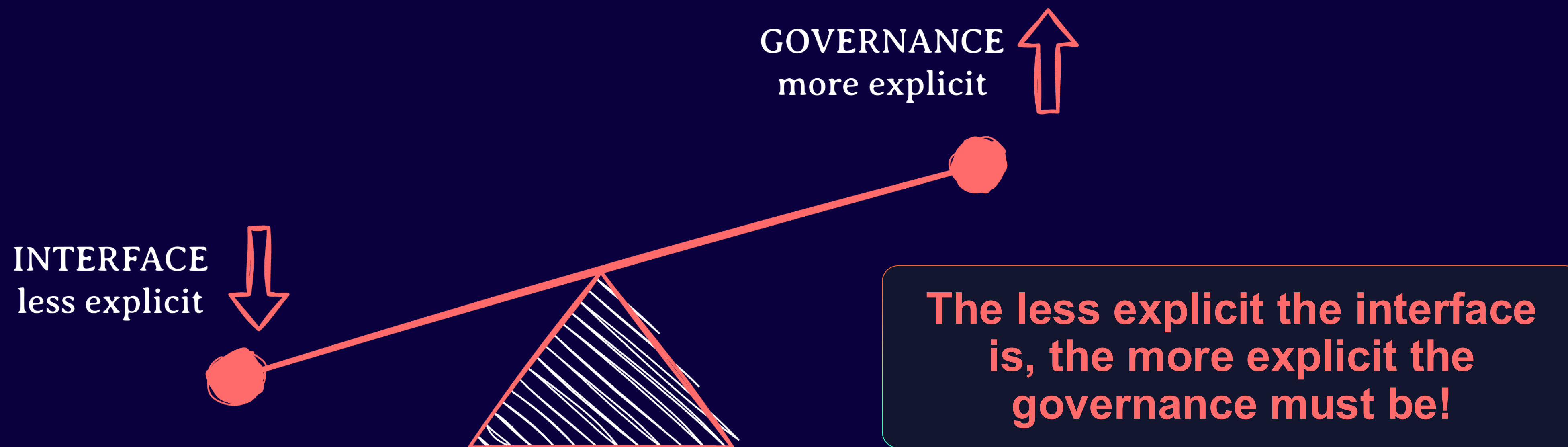
Sender Provisioning



ACT IV:

FAILURE MODES & DESIGN RULES

Interface and Governance



The lethal trifecta

Access to private data

**Exposure to untrusted
content**

**Ability to externally
communicate**

**Never let your agent combine
these three features**

Some design rules

- Deterministic delivery stays on APIs
- Separate read/write tools
- Confirm destructive/expensive actions
- Scope by tenant/role
- Log every tool call
- Use structured output

**Most of this is stuff
you already know –
just applied to a
new caller!**

Don't wrap every endpoint

- More tools $\neq$ Better outcome
- If we expose everything as tools, we get:
 - Tool bloat
 - Context rot
 - Worse agents

The way to do it: Curate and design **task-oriented tools**

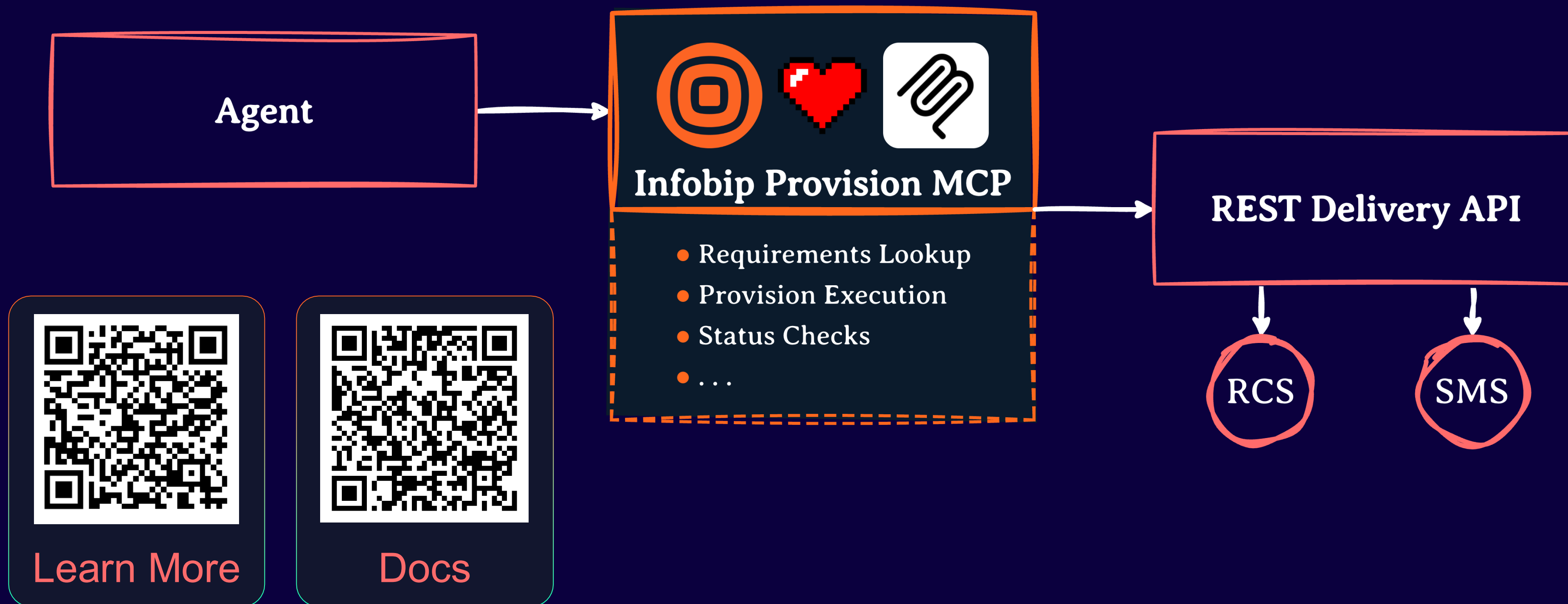
ACT V: *PROOF*

This is how we do



All these companies kept their transactional APIs, and **added MCP for ops and dev-assist**

Infobip Provisioning MCP server



ACT VI: *CONCLUSION*

Takeaways

- Keep deterministic delivery on explicit contracts
- Use MCP for the operational workflows around them
- The less explicit the interface, the more explicit the governance

Takeaways

REST is how the system does the work.

MCP is how agents understand, configure, and operate it!

ACT VII:

Q&A

References

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Thank you!

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